Clock and acceleration predictions of a fixed scalar–tensor branch: calibration and Yukawa constraints

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preliminary research; numerical cross-checks with Academia Co-scientist (AI-assisted)

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Abstract

We fix one branch of the TAL scalar–tensor line (V26 Branch II-Fluid with universal matter coupling, assumption U) and three declared parameter sets, P1 $(L, \alpha_Y) = (10\,\mathrm{km}, 1)$, P2 $(10\,\mathrm{km}, 1/3)$ and P3 $(1\,\mathrm{km}, 1)$. From the action we derive, for a declared granite source moved between two positions, the change of the two-clock frequency ratio and of the horizontal acceleration at the clocks, with the Newton constant calibrated in the laboratory and a GR reference that uses the same source and the same G_{lab}. No parameter is fitted. For P1 the calibrated differences are $\delta y = 1.432 \times 10^{-34}$ (dimensionless) and $\delta a = 7.355 \times 10^{-18}\,\mathrm{m\,s^{-2}}$. The Yukawa force of all three sets lies above every examined bound: the conditional ratio to the bound is $R \geq 3.19$ (BREN tower, single-point comparisons within the examined source and operator models) and $R \geq 58.55$ (laboratory compilation). R is a ratio to a bound, not a significance. The fixed branch is therefore in category (b): a signal exists, but its force is excluded by the examined bounds. For $\alpha_Y = 1$ the part of the family that the examined sub-centimetre bounds do not exclude ($\lambda < 75.877\,\mu\mathrm{m}$ with the drawn $|\alpha|$ curves, up to $101.029\,\mu\mathrm{m}$ with the sign-separated $+\alpha_{95}$ construction) is category (c): the computed signal is nonzero but far below the declared resolution. The BREN part is a project recomputation with synthetic ring radii, a modern substitute DEM and uniform density; it is not an empirical reanalysis of the original experiment.

1 Introduction and scope

This note combines three runnable modules of one publication set: the clock and acceleration analysis of the fixed branch (the main analysis) and two bound modules, the BREN tower continuation and the sub-centimetre Yukawa window. The clock module reads its bound inputs from the other two, so the three modules are not independent evidence; they are one chain. Every number in the text and in the tables is read by a script from the module outputs and references (Section 4 and the claim–source table `CLAIMS_SOURCES.csv`).

We use four decision categories: (a) no new signal; (b) a signal exists, but its force is excluded by existing bounds; (c) the signal is allowed by the examined bounds, but too small to resolve with the declared uncertainty proxies; (d) allowed and resolvable, i.e. a test that could be pre-registered. "Not excluded" means category (c) at most; it does not mean confirmed, and a computed nonzero signal is not a measured signal.

2 Action, matter coupling and notation

In units $G = c = 1$ the branch action is

$$S = \int d^4x\,\sqrt{-g}\Big[\frac{R}{16\pi} - \frac{1}{2\alpha_Y}\Big((\partial\phi)^2 + \frac{\phi^2}{L^2}\Big) - f(\phi)\,\varepsilon(n)\Big], \qquad f(\phi) = e^{-\phi}, \tag{1}$$

with one metric g and the rest-frame energy density $\varepsilon(n)$ of the matter fluid; the field equation is $\Box\phi - \phi/L^2 = -\alpha_Y e^{-\phi}\varepsilon$ (TAL V26, `10.5281/zenodo.23727585`). Three couplings are kept

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apart. α_Y is the coupling in the action. The Yukawa strength in the standard convention $V = -G_\infty m_1 m_2 r^{-1}(1 + \alpha e^{-r/\lambda})$ is $\alpha = \alpha_Y/(4\pi)$, with range $\lambda = L$. For $\alpha_Y = 1$ this is $q = 1/(4\pi) = 0.0795775$, the value compared with the bounds below.

Assumption U. Every non-gravitational field (clock atoms, quartz, electronics, test masses) couples to the matter metric $\tilde{g}_{\mu\nu} = f^2(\phi) g_{\mu\nu}$. This is a declared part of the branch, not a derived result. A clock-only coupling (the H1 form) and an anomalous clock sensitivity k_c are outside the branch; they are kept separate (Appendix C). The long-range MICROSCOPE proxy for k_c is not transferred to $L = 1$–$10\,\mathrm{km}$.

Parameter sets (declared, not fitted). Table 1 lists the three sets; α' and $G_\infty/G_{\rm lab}$ are defined in Section 3.

Table 1: Declared parameter sets.

set	L (km)	α_Y	$\alpha = \alpha_Y/(4\pi)$	α'	$G_\infty/G_{\rm lab}$
P1	10	1	0.07958	0.07371	0.926288
P2	10	1/3	0.02653	0.02584	0.974160
P3	1	1	0.07958	0.07371	0.926288

3 Clock and acceleration from one source

3.1 One potential for both observables

In the static weak field the matter metric has $\tilde{g}_{tt} = -e^{2U/c^2}$ with

$$U(\mathbf{x}) = -G_\infty \int \frac{\rho(\mathbf{x}')}{d}(1 + \alpha e^{-d/L}) d^3x', \qquad d = |\mathbf{x} - \mathbf{x}'|. \tag{2}$$

Under U each clock ticks in $\tilde{g}$ proper time; the comparison signals obey conformally invariant Maxwell equations, so for two identical clocks

$$y \equiv \ln \frac{\nu_A}{\nu_B} = \frac{U(\mathbf{x}_A) - U(\mathbf{x}_B)}{c^2}, \qquad \mathbf{a} = -\nabla U. \tag{3}$$

The frequency ratio and $1 + z$ are reciprocal, $\ln(1 + z) = -\Delta \ln \nu$; we report $\ln \nu$. The redshift and the force come from the same U, so under U the clock carries no information independent of the force.

3.2 Laboratory calibration of G

A laboratory measurement of the Newton constant at an effective separation s returns

$$G_{\rm lab} = G_\infty\big[1 + \alpha\,\kappa(s_{\rm cal}/L)\big], \qquad \kappa(x) = e^{-x}(1 + x), \tag{4}$$

with declared $s_{\rm cal} = 0.10\,\mathrm{m}$ and $G_{\rm lab} = 6.6743 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$ (CODATA 2018). For $L \gg s_{\rm cal}$ this gives $G_\infty = G_{\rm lab}/(1 + \alpha)$, the convention of TAL V28 Step 27 (`10.5281/zenodo.23814053`). The GR model of the laboratory uses the same source and $G_{\rm lab}$: $U_{\rm GR} = -G_{\rm lab} \int \rho/d\, d^3x'$. The difference is

$$\delta U = -G_{\rm lab}\,\alpha' \int \rho\, \frac{e^{-d/L} - \kappa(s_{\rm cal}/L)}{d}\, d^3x', \qquad \alpha' = \frac{\alpha}{1 + \alpha\kappa(s_{\rm cal}/L)}. \tag{5}$$

The sensitivity variants $s_{\rm cal} = 0.05$ and $0.20\,\mathrm{m}$ are computed in every run of the clock module (they are part of its reference results), so no separate run was needed; Table 3 gives them. A raw column (the Yukawa term alone with $G_{\rm lab}$, no calibration) is kept in the module only as an uncalibrated illustration; for P1 it is $\delta y_{\rm raw} = -4.97 \times 10^{-26}$ and $\delta a_{\rm raw} = -7.79 \times 10^{-9}\,\mathrm{m\,s^{-2}}$. It is not the observable.

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3.3 Observables

With S1 and S2 the two source positions and a_x the component along the baseline $A \to B$,

$$\delta y = \left[y(S1) - y(S2)\right]_{\mathrm{TAL}} - \left[y(S1) - y(S2)\right]_{\mathrm{GR}}, \tag{6}$$

$$\delta a = \left[a_x^A(S1) - a_x^A(S2)\right]_{\mathrm{TAL}} - \left[a_x^A(S1) - a_x^A(S2)\right]_{\mathrm{GR}}. \tag{7}$$

The order is S1−S2 and TAL−GR throughout. δy is dimensionless; δa is in $\mathrm{m\,s^{-2}}$. The Earth's potential and scalar field are the same for S1 and S2 and cancel.

3.4 Declared geometry

The source is a uniform granite cylindrical shell with $r = 0.09$–$0.22\,\mathrm{m}$, height $0.50\,\mathrm{m}$, $\rho = 2700\,\mathrm{kg\,m^{-3}}$ and mass $170.9\,\mathrm{kg}$ (TAL V20, `10.5281/zenodo.23540741`). Clock A is at the origin and clock B at $1.60\,\mathrm{m}$ on the same height. S1 puts the source axis at $x = -0.32\,\mathrm{m}$, S2 at $x = 1.92\,\mathrm{m}$, so the outer surface is $0.10\,\mathrm{m}$ from the nearer clock; this is the minimum distance between any clock and the source mass. The geometry is declared: the plan files of the original rule were not available, so the distance rule is stated as an assumption. The results hold for this declared geometry only; no other source-mass model is used. The GR reference values are $\Delta y_{\mathrm{GR}} = -6.247 \times 10^{-25}$ and $\Delta a_{\mathrm{GR}} = -9.793 \times 10^{-8}\,\mathrm{m\,s^{-2}}$.

4 Numerical method, checks and reproducibility

4.1 Clock module

The potential and the force are integrated by panel Gauss–Legendre quadrature in (r, z, φ) with 12 nodes per panel, for $10^{-2} \le L \le 10^{5}\,\mathrm{m}$ (8 points per decade). Below $L = 10^{-2}\,\mathrm{m}$ only an analytic upper bound is evaluated (any source with density $\le \rho$ and all mass at distance $\ge 0.10\,\mathrm{m}$), in $\log_{10}$, down to $L = 10^{-5}\,\mathrm{m}$ with 8 points per decade; it is neither a quadrature result nor a numerical zero. The conclusion for $\lambda < 10^{-2}\,\mathrm{m}$ is therefore an upper bound for the declared geometry, whose closest mass is $0.10\,\mathrm{m}$ away. `run_check.py` verifies the input SHA-256 sums and the provenance of every bound file, runs 16 physics self-tests (mass, closed-form limits, the calibration limit, redshift equal to the line integral of the force, swap symmetry, grid refinement, the analytic bound above the quadrature, the BREN and sub-cm inputs) and 5 deliberately broken controls that must fail, among them non-finite values, a perturbed reference, a one-byte input change and a wrong provenance record. Outputs are compared with the reference results within a relative tolerance of 10^{-8}.

4.2 Bound modules

The BREN module checks the operator against brute-force and analytic cases (prism 8.7×10^{-6}, Yukawa near field 5.2×10^{-5}, harmonic point mass 1.8×10^{-6} relative deviation), reproduces the V28 Step 27 numbers exactly (largest relative difference 0), and runs `check_station_surfaces.py` for the base and 10 further configurations (11 in total, 0 mismatches between station heights and source-prism surfaces). The sub-cm module checks its digitized curves against 17 values stated in the sources (17 consistent), e.g. the Eöt-Wash 2020 curve gives $38.57\,\mu\mathrm{m}$ at $|\alpha| = 1$ against the stated $38.6\,\mu\mathrm{m}$.

4.3 Reproducibility

The basis is the Zenodo variant of each module; third-party PDFs are not redistributed, and the CC BY-NC licence of the modules does not extend to third-party documents and data. The three archives are unpacked into three separate directories, never on top of each other. For this note each module was run once from a fresh unpacking of its archive (logs in `logs/`): `python3 run_check.py --out runs/fresh_article` (clock; `RESULT: PASS`), `python3 run_check.py` (BREN; "all steps passed"),

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`python3 run_check.py` (sub-cm; "all steps passed (frozen-digitization mode)"), with Python 3.13.5, NumPy 2.5.4, SciPy 1.18.1, matplotlib 3.11.2 and Pillow 13.0.0 on Linux. `requirements.txt` records the versions used; it does not by itself prove an environment. Figures are not expected to be byte-identical; numbers are checked against the references within tolerance. The sub-cm base run starts from the frozen digitization data (no new PDF digitization). Regenerating the BREN DEM from the raw Copernicus tiles is a separate task that the module does not perform. The four bound files that the clock module reads are byte-identical to the upstream module outputs and match `PROVENANCE_bounds.json`; the file name `bren_continuation_results_r2.json` is historical. The comparison scripts treat NaN and ±Inf as differences, and the negative controls stay in the suites.

5 Results for P1–P3

Table 2 gives the calibrated differences. Relative to the GR source effect, $\delta y/\Delta y_{\mathrm{GR}}$ runs from -2.29×10^{-10} (P1) to -2.29×10^{-8} (P3). The dependence on the declared calibration separation is shown in Table 3.

Table 2: Calibrated TAL−GR differences (S1−S2). δy dimensionless, δa in $\mathrm{m\,s^{-2}}$.

set	δy	δa ($\mathrm{m\,s^{-2}}$)	$\delta y/\Delta y_{\mathrm{GR}}$	$\delta a/\Delta a_{\mathrm{GR}}$
P1	1.432×10^{-34}	7.355×10^{-18}	-2.29×10^{-10}	-7.51×10^{-11}
P2	5.020×10^{-35}	2.579×10^{-18}	-8.04×10^{-11}	-2.63×10^{-11}
P3	1.431×10^{-32}	7.350×10^{-16}	-2.29×10^{-8}	-7.51×10^{-9}

Table 3: Sensitivity to the calibration separation s_{cal} (declared value in the middle column of each group).

	δy			δa ($\mathrm{m\,s^{-2}}$)		
set	$s_{\mathrm{cal}} = 0.05\,\mathrm{m}$	$0.10\,\mathrm{m}$	$0.20\,\mathrm{m}$	$0.05\,\mathrm{m}$	$0.10\,\mathrm{m}$	$0.20\,\mathrm{m}$
P1	1.449×10^{-34}	1.432×10^{-34}	1.363×10^{-34}	7.626×10^{-18}	7.355×10^{-18}	6.273×10^{-18}
P2	5.081×10^{-35}	5.020×10^{-35}	4.778×10^{-35}	2.673×10^{-18}	2.579×10^{-18}	2.199×10^{-18}
P3	1.448×10^{-32}	1.431×10^{-32}	1.362×10^{-32}	7.620×10^{-16}	7.350×10^{-16}	6.267×10^{-16}

Feasibility. We compare the signals with declared proxies: OCXO 8663 short-term stability 1×10^{-11} (the measured mean beat offset 2.56×10^{-11} is an offset, not a noise floor), the Sr single-clock systematic 8.1×10^{-19} [Aeppli 2024], the Sr differential 7.6×10^{-21} [Bothwell 2022] and a torsion-balance differential acceleration $3.1 \times 10^{-15}\,\mathrm{m\,s^{-2}}$ [Schlamminger 2008]. These come from different protocols. The GR model of the source carries a relative uncertainty $\epsilon = 0.0185$ (a proxy from two granite densities) or 1×10^{-4} (a declared best case). $S = |\delta|/\sqrt{\sigma_{\mathrm{inst}}^2 + (\epsilon\,\Delta_{\mathrm{GR}})^2}$ is a feasibility indicator built from these proxies, not the significance of an observed signal; a scenario would count as resolvable at $S = 5$ (Table 4). The largest S is 7.51×10^{-5} (P3); the signal would have to be 6.662×10^{4} times larger to reach $S = 5$ in the best declared scenario.

Table 4: Feasibility indicator S (not a significance).

set	S_y, OCXO $\epsilon = 0.0185$	S_y, Sr diff. $\epsilon = 1 \times 10^{-4}$	S_a, torsion $\epsilon = 0.0185$	S_a, torsion $\epsilon = 1 \times 10^{-4}$	largest S
P1	1.4×10^{-23}	1.9×10^{-14}	4.1×10^{-9}	7.5×10^{-7}	7.51×10^{-7}
P2	5.0×10^{-24}	6.6×10^{-15}	1.4×10^{-9}	2.6×10^{-7}	2.63×10^{-7}
P3	1.4×10^{-21}	1.9×10^{-12}	4.1×10^{-7}	7.5×10^{-5}	7.51×10^{-5}

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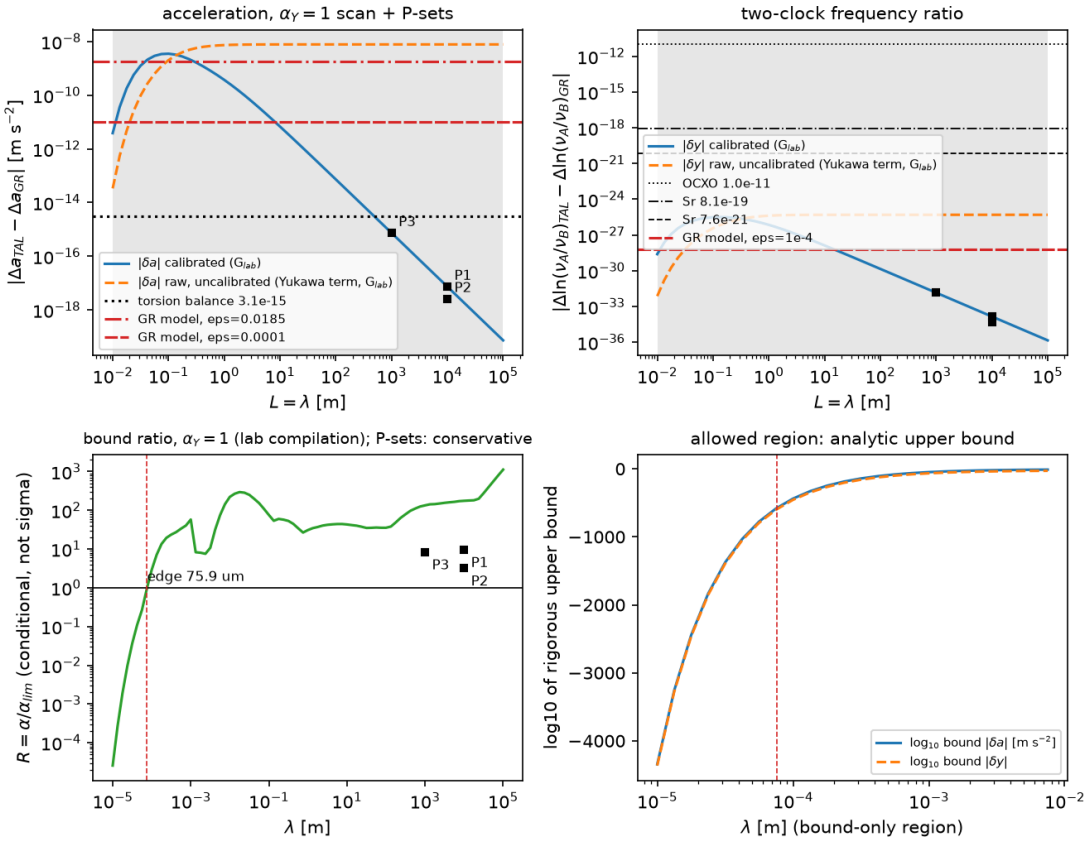


Figure 1: Clock module figure: $|\delta a|$ and $|\delta y|$ against L for $\alpha_Y = 1$ and P1–P3; the ratio R; and, below 10^{-2} m, the analytic upper bound. The dashed raw curves are an uncalibrated illustration.

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6 Yukawa bounds

6.1 BREN tower chain

The BREN module applies the continuation operator of the tower reports [Kasameyer 1989; Kammeraad 1990] to one source model and compares the tower residual of a Yukawa term with the flat half-space form of V28 Step 27, at the model point $\alpha = 0.0795775$, $\lambda = 10\,\mathrm{km}$. The source model uses synthetic ring radii (45 rings, 1838 stations; the reports give counts but not radii), a modern substitute DEM (Copernicus GLO-90, 301×301 cells of 180 m) and uniform density. The original station and data chain is not reconstructed, so this is not an empirical reanalysis of the experiment. The observable is referenced to $z_0 = 9.8\,\mathrm{m}$ (K90); the V28 Step 27 numbers used $z_0 = 0$ and are kept separate ($R = 10.531$ at $\rho = 2000\,\mathrm{kg\,m^{-3}}$ and 14.085 at $2670\,\mathrm{kg\,m^{-3}}$, reproduced exactly).

Variants: A, spherical kernel with the raw truncated weights (sum -0.00197 with truncation at 240 km); B, weights renormalised to zero sum; C, planar kernel; D, as A with a linear fit of the Newtonian bias. The truncation sensitivity of A is an operator property, not a terrain signal, and the spread between A and B is not an uncertainty band. At the top point (457.15 m, $\rho = 2200\,\mathrm{kg\,m^{-3}}$) the flat residual is $-2972.3\,\mu$Gal; A gives $-2766.9\,\mu$Gal, B gives $-2900.6\,\mu$Gal; the operator-only part of A is $133.9\,\mu$Gal; the terrain part, common to all variants, is $71.5\,\mu$Gal (2.4 % of the flat value). As an independent check, the Newtonian terrain bias (direct minus continued) is $137.6\,\mu$Gal against the report's $140 \pm 20\,\mu$Gal, and $111.9\,\mu$Gal at 454 m with $\rho = 1800\,\mathrm{kg\,m^{-3}}$ against about $100\,\mu$Gal in K89 (not tuned).

Two conditional single-point comparisons are used. B89: the K89 residual $93 \pm 95\,\mu$Gal at the 454 m top, taken with the conservative sign and $k = 1.96$, gives an edge of $-279.2\,\mu$Gal. B90: the digitized K90 Fig. 1 method-1 top point $+3.5 \pm 40.3\,\mu$Gal at 457.15 m (tower height 465 m) gives $-75.5\,\mu$Gal. Both use only the variance of one pre-selected point; the off-diagonal covariance with other heights does not enter a single-point marginal test, which `check_single_point_marginal.py` confirms (7 of 7 covariance structures leave the edge unchanged). The K90 correlated χ^2 analysis is not reproduced. Table 5 gives the ratios at $\lambda = 10\,\mathrm{km}$ with density, height and construction per row.

Table 5: BREN: $R = \alpha/\alpha_{\mathrm{lim}}$ at $\lambda = 10\,\mathrm{km}$, $z_0 = 9.8\,\mathrm{m}$, $k = 1.96$. B89: K89 top point, 454 m; B90: K90 Fig. 1 top point, 457.15 m (conditional single-point comparisons; not σ).

ρ ($\mathrm{kg\,m^{-3}}$)	bound	R flat	R A	R B	R C	R D
2000	B89	10.297	9.580	10.046	9.605	9.580
2000	B90	38.564	35.894	37.633	35.986	35.896
2200	B89	11.334	10.546	11.059	10.573	10.546
2200	B90	42.429	39.491	41.404	39.593	39.494
2670	B89	13.773	12.815	13.439	12.849	12.816
2670	B90	51.510	47.945	50.266	48.069	47.948

The DEM preparation uses a local equirectangular mapping; its east–west metric changes by up to 0.32 % at the north and south edges. This is a declared limitation; it is not corrected, and it is not automatically a residual error of that size.

6.2 Sub-centimetre chain

The sub-cm module compares $\alpha = 0.0795775$ with published short-range bounds (95 % CL), vector-digitized from Adelberger et al. 2003 and Lee et al. 2020, and with Table 11.4 of the Lee 2020 thesis (66 rows). The edges of the transition band are $75.877\,\mu$m (HUST 2020 $|\alpha|$ curve, reading interval 74.71–77.06 μm), $98.074\,\mu$m (Eöt-Wash 2020 $|\alpha|$ curve, 95.61–100.60 μm), $98.136\,\mu$m (thesis $|\alpha_{95}|$, 98.07–98.31 μm) and $101.029\,\mu$m (thesis $+\alpha_{95}$, 100.99–101.12 μm). The band is not only a HUST sign question. From 75.877 to about $98.074\,\mu$m, α is excluded only if the HUST 2020/2016 $|\alpha|$

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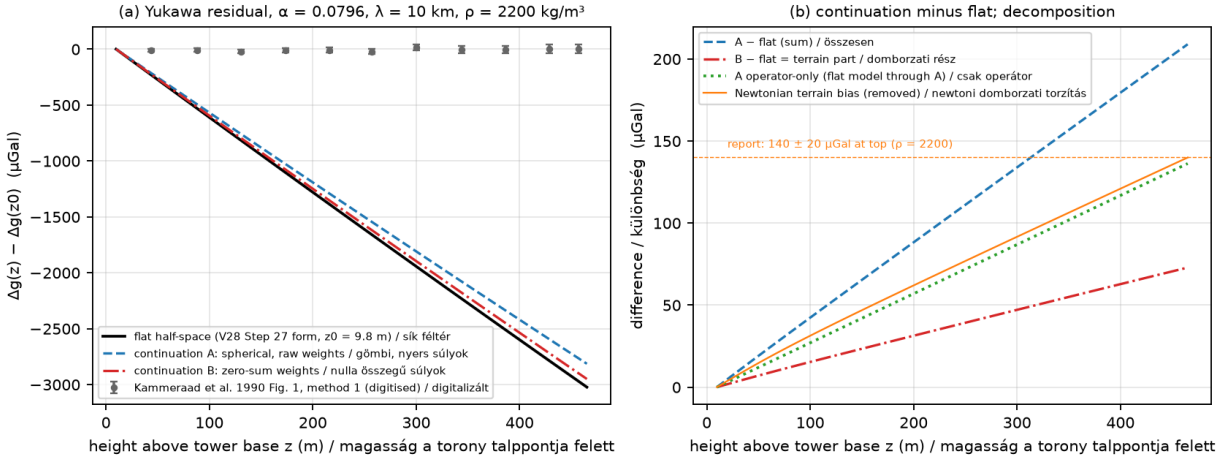


Figure 2: BREN module figure: (a) Yukawa residual against height for the flat half-space and variants A and B, with the digitized K90 points; (b) continuation minus flat, decomposed.

curves are applied to $\alpha > 0$. From about 98.074 to 101.029 μm the two Eöt-Wash 2020 constructions differ without any HUST curve: the thesis defines $\pm\alpha_{95}$ from the $\Delta\chi^2 = 6.17$ surface and $|\alpha_{95}|$ from a normal integral; in the 100 μm row, $+\alpha_{95} = 0.0813 > \alpha > |\alpha_{95}| = 0.0766$. The bounds are used as published. The reading intervals come from the digitization and are not a physical uncertainty. In the range $\lambda < 75.877\,\mu$m the value $\alpha = q$ is not excluded by the examined bounds; the search for newer bounds was not exhaustive.

Table 6: Sub-cm: $R = \alpha/\alpha_{\rm lim}$ at selected λ (conditional, not σ).

λ (μm)	R, thesis $+\alpha_{95}$	R, drawn $\lvert\alpha\rvert$ envelope
50	0.161	0.190
100	0.979	2.92
200	3.25	16.0
1000	8.81	57.8

6.3 Ratios for P1–P3

$R = \alpha/\alpha_{\rm lim}$ is a conditional ratio to a bound, not a significance: R is not a number of standard deviations, and the bounds are at different levels ($k = 1.96$ or 95 % CL for the laboratory and BREN single-point limits, 1σ for the quoted K90 limit). Table 7 combines the laboratory compilation (Adelberger et al. 2003 Fig. 4 above 2.00 mm with the sub-cm envelope) and the BREN B89 limits at $\rho = 2000\,\mathrm{kg\,m^{-3}}$ (variants A–C). For P3 the K90 report limit 0.0032 at 1 km gives $R = 24.9$; the project's B90 single-point value 1.13×10^{-3} ($R = 70.7$) is a different construction, context only, and does not enter the conservative ratio. Under U the redshift tests [Takamoto 2020; Grotti 2018] and the MICROSCOPE test [Touboul 2022] do not constrain the branch.

Table 7: Bound ratios (conditional, not σ) and categories.

set	$\alpha_{\rm lim}$ lab	R lab	$\alpha_{\rm lim}$ BREN (A, B, C)	R BREN	R conservative	category
P1	4.53×10^{-4}	175.6	8.31×10^{-3}, 7.92×10^{-3}, 8.29×10^{-3}	9.58–10.05	9.58	(b)
P2	4.53×10^{-4}	58.55	8.31×10^{-3}, 7.92×10^{-3}, 8.29×10^{-3}	3.19–3.35	3.19	(b)
P3	5.94×10^{-4}	134.0	9.52×10^{-3}, 9.47×10^{-3}, 9.52×10^{-3}	8.36–8.40	8.36	(b)

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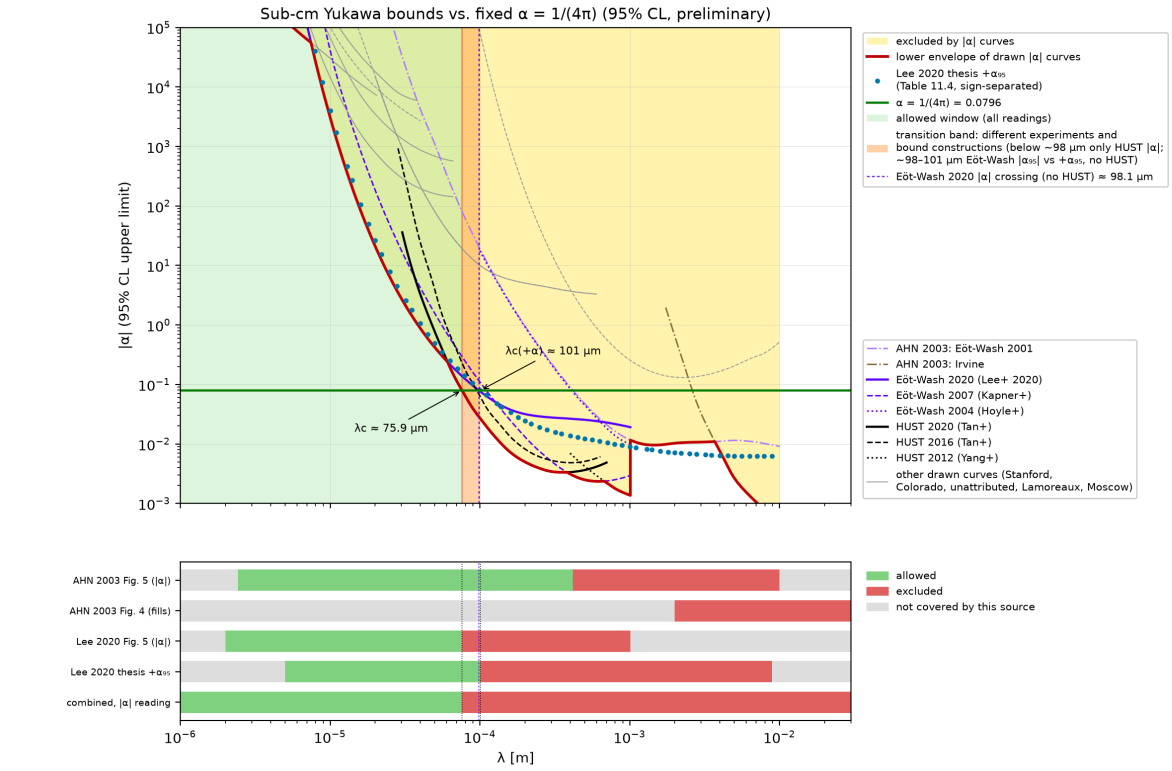


Figure 3: Sub-cm module figure: drawn $|\alpha|$ bounds, the thesis $+\alpha_{95}$ points and $\alpha = 1/(4\pi)$, with the transition band.

7 Conclusion

- The fixed branch (P1, P2, P3) is in category (b): $\delta y \neq 0$ and $\delta a \neq 0$, but the force is excluded by the examined bounds, with smallest ratios $R = 9.58$, 3.19, 8.36 (BREN, conditional on the examined models).
- For $\alpha_Y = 1$ the range $\lambda < 75.88\,\mu$m is not excluded by the examined bounds and is category (c): there $\log_{10}|\delta a/(\mathrm{m\,s^{-2}})| \leq -576.0$ and $\log_{10}|\delta y| \leq -593.6$. In the widest reading, up to $101.029\,\mu$m, the bound gives $\log_{10}|\delta a/(\mathrm{m\,s^{-2}})| \leq -433.6$ and $\log_{10}|\delta y| \leq -451.3$, category (c). This conclusion is for the declared geometry, whose closest mass is 0.10 m from the clocks.
- No category (d) test exists for this branch in this setup. A short-range force test could probe $\alpha_Y = 1$: at $\lambda = 50\,\mu$m the drawn envelope gives $R = 0.190$, so a torsion balance about 5.3 times more sensitive there would reach $R = 1$. This is a force experiment, not a clock experiment.

The checks reported here are technical (scripted reproduction, numeric comparison, negative controls); a passing check is not an empirical confirmation of the branch. They were done as numerical cross-checks with Academia Co-scientist (AI-assisted).

A BREN operator

The operator of K89 eq. 7 (= K90 eq. 2) is the spherical Poisson expansion to order z/a,

$$\Delta g(z) = \frac{z(1-2z/a)}{2\pi} \iint \frac{\Delta g(0,r',\varphi)\, r'\, dr'\, d\varphi}{\left[r'^2 + z^2(1-z/a)\right]^{3/2}}, \qquad a = 6371\,\mathrm{km}, \tag{8}$$

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discretised into rings; ring i (radii r_1, r_2) carries the mean of its sector stations times $K_i(z) = z(1-2z/a)\big[(r_1^2+b^2)^{-1/2} - (r_2^2+b^2)^{-1/2}\big]$, $b^2 = z^2(1-z/a)$, truncated at 240 km. With T the direct tower value and P the continued value, $B_X(z) = [T_X(z) - T_X(z_0)] - [P_X(z) - P_X(z_0)]$ for Newton ($X = N$) and Yukawa ($X = Y$), and the Yukawa residual is $\Delta g = G_\infty \alpha \rho\, (B_Y - B_N)$; in the flat limit it returns the Step 27 form. Tower location 36.78059° N, 116.24359° W; 103728 signed terrain prisms. Convergence at the top ($\rho = 2200\,\mathrm{kg\,m^{-3}}$, 10 configurations besides the base): variant B changes by at most 2.44 μGal; variant A runs from $-2388.0\,\mu$Gal at $R_{\mathrm{trunc}} = 60$ km to $-2829.9\,\mu$Gal at 480 km, as expected for the truncation term; the Newtonian bias spans 107.4–165.6 μGal and cancels in the Yukawa residual. Table 8 shows α_{lim} against λ (B89, $\rho = 2000\,\mathrm{kg\,m^{-3}}$).

Table 8: BREN: α_{lim} against λ (B89, $\rho = 2000\,\mathrm{kg\,m^{-3}}$).

λ (km)	flat	A	B
0.300	0.0151	0.0150	0.0150
1.00	0.00946	0.00952	0.00947
3.00	0.00815	0.00838	0.00826
10.0	0.00773	0.00831	0.00792
50.0	0.00759	0.0101	0.00781

B Digitization and bound constructions

The sub-cm curves were digitized from the vector content of the published figures (stroke paths, axis calibration by a fit to all tick marks) and checked against a raster reading; the thesis table was parsed from the text layer. The digitized values are frozen in the module, with SHA-256 sums, and are the input of its default run. The constructions differ: the drawn $|\alpha|$ curves are bounds on $|\alpha|$; the thesis $\pm\alpha_{95}$ come from the $\Delta\chi^2 = 6.17$ surface; $|\alpha_{95}|$ comes from the normal integral between $-|\alpha_{95}|$ and $+|\alpha_{95}|$. The older AHN 2003 Fig. 5 envelope alone would put the edge at 417 μm; the HUST 2016 curve crosses at 95.27 μm; the HUST-free gap is 2.89 μm wide. For BREN, B89 uses a value stated in K89 with $k = 1.96$, and B90 a point digitized from K90 Fig. 1; the K90 limit itself (1σ, global multi-height correlated χ^2) is quoted, not recomputed.

C Non-universal addon (outside the fixed branch)

This appendix is an illustration outside the fixed branch and does not enter the decision. If the clock frequency had an anomalous sensitivity, $\nu \propto e^{-(1+k_c)\phi}$, then $\delta y_{\mathrm{extra}} = -k_c\,\Delta(\phi_A - \phi_B)$, with ϕ normalized by G_∞ (corrected in module version r4). Table 9 gives $\delta y_{\mathrm{extra}}$ for $k_c = 1$ and the k_c needed for $S = 5$ at the Sr differential level. A long-range proxy with the Coulomb-energy fractions $\xi_{\mathrm{Ti}} = 2.022 \times 10^{-3}$, $\xi_{\mathrm{Pt}} = 4.054 \times 10^{-3}$ and MICROSCOPE $|\eta| < 6.882 \times 10^{-15}$ gives $|k_c| \lesssim 4.255 \times 10^{-11}$; this proxy is not transferred to $L = 1$–10 km.

Table 9: Non-universal addon (illustration only).

set	$G_\infty/G_{\mathrm{lab}}$	$\delta y_{\mathrm{extra}}$ ($k_c = 1$)	k_c for $S = 5$
P1	0.926288	-4.605×10^{-26}	8.252×10^{5}
P2	0.974160	-1.614×10^{-26}	2.354×10^{6}
P3	0.926288	-4.605×10^{-26}	8.252×10^{5}

Data and code availability

The three module archives are in the publication set, unchanged, with SHA-256 sums:

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- `tal_branch_clock_accel_test_r4.zip` (clock and acceleration, main analysis);
- `bren_continuation_zenodo_r4.zip` (BREN tower continuation);
- `subcm_yukawa_window_zenodo_r4.zip` (sub-centimetre Yukawa window).

The article sources, the claim–source table and its check script are in `article/`. Concept DOI of the TAL work: `10.5281/zenodo.22983681`; related records: V20 `10.5281/zenodo.23540741`, V26 `10.5281/zenodo.23727585`, V28 `10.5281/zenodo.23814053`. Licence: CC BY-NC 4.0 International; third-party documents and data remain under their own terms.

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